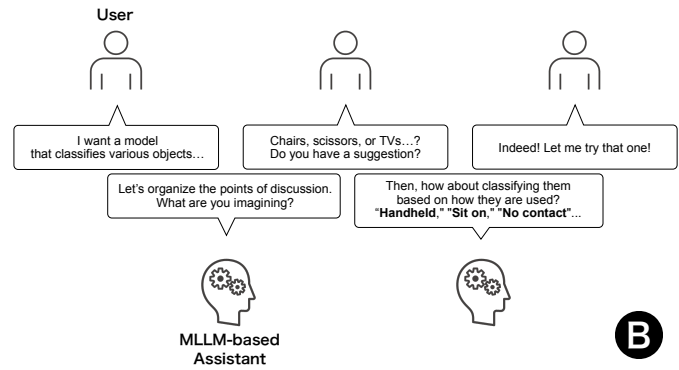
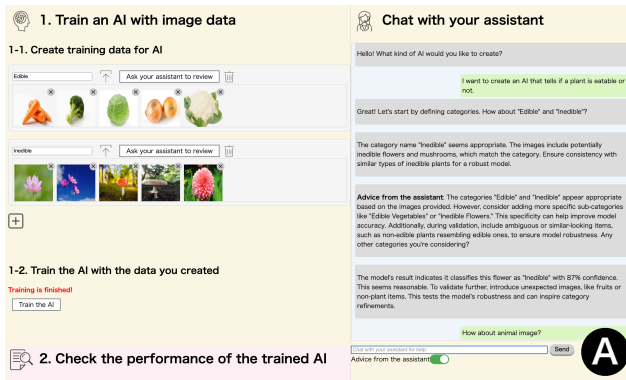


# A Multimodal LLM-based Assistant for User-Centric Interactive Machine Learning

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**Figure 1: (A) Our system aims to assist users without a technical background in appropriately formulating tasks and creating training data in machine learning prototyping. (B) The system uses a multimodal large language model (MLLM)-based assistant to elicit user needs and guide them interactively toward appropriate training data.**

## Abstract

This paper proposes a system based on a multimodal large language model (MLLM) to assist non-expert users without prior experience in machine learning (ML) development. The MLLM assistant in our system interactively helps users compile their requirements and create appropriate training data while building an ML model. It has been reported that users often struggle to define training data that comprehensively covers all samples or aligns with their needs. To prevent such failures, the MLLM assistant monitors the user's interaction process and translates users' vague needs into concrete ML formulations through chat, ultimately facilitating the creation of appropriate training data.

## CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI); Interaction design;** • **Information systems;**

## Keywords

graphical user interface, machine learning, large language model

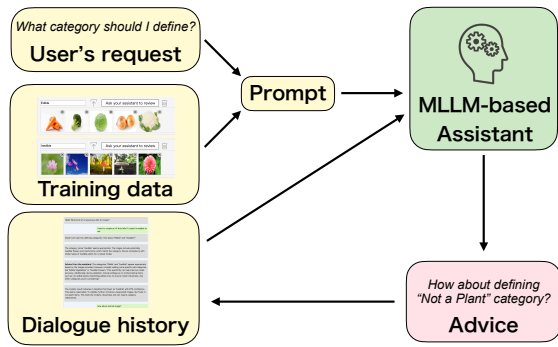
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## 1 Introduction

Interactive Machine Learning (IML) is a framework designed to enable simplified ML development through UI operations [Amer-shi et al. 2014; Dudley and Kristensson 2018]. IML aims to allow non-expert users without a technical background to engage in ML development. However, it has been reported that developing models to solve tasks using IML appropriately is not always straightforward for users [Kawabe and Sugano 2024; Nakao and Sugano 2020]. In the context of image classification, for instance, users may fail to enumerate all the necessary categories or may define categories that do not align with the intended task when defining training data. This difficulty arises from the gap between the task users want to address and the way it is concretely formulated as an ML problem and written as training data.

To bridge this gap, users need to realize how to appropriately formulate their needs into tasks. For example, if users want a model to determine whether a plant is edible, the categories should not be specific names like "Tomato" or "Mushroom" but rather abstract categories like "Edible", "Non-Edible", and "Non-Plant". Novice users often struggle to reach this understanding on their own. On the other hand, recent multimodal large language models (MLLMs) can perform such task formulations and image recognition in a very generalized manner. However, it is still difficult to create a system



**Figure 2: Data processing flow. The assistant generates advice by comprehensively referencing information.**

that automatically understands the user’s demands, and it is also a challenging task for novice users to give appropriate prompts to MLLMs. The key idea of our study is to design a system where, within the IML process, users and the MLLM work together, extracting mutual knowledge through collaboration to achieve appropriate training data.

We propose a system that integrates the chat capabilities of a MLLM into existing IML systems, leveraging their strengths within the IML framework. Simply providing an MLLM as it is might prevent users from fully articulating their underlying ideas. Therefore, we designed an interaction where the MLLM monitors the entire IML process and acts as an assistant, interpreting users’ needs and thinking alongside them. The role of the MLLM within the system is to guide users’ vague needs into the creation of appropriate training data through chat interactions.

## 2 PROPOSED SYSTEM

Figure 1 (A) shows the appearance of the GUI for our proposed system. The GUI is primarily divided into two areas: the IML part and the chat part. In the IML part, users follow the basic ML development flow: defining training data, training the model with the training data, and evaluating its performance. Throughout this process, an MLLM-based assistant monitors the entire workflow and provides advice to the user (Fig. 1 (B)) in the chat part.

Our system provides users with multiple ways to collaborate with an MLLM assistant. Users may (1) actively seek advice when they have questions or (2) not actively seek advice, simply not thinking to ask for guidance at that moment. (1) If users want to ask the assistant something, they can use chat input (e.g., User: “I want to create a model that can determine if a plant is edible from its image. What kind of training data should I prepare?”). Additionally, at the locations where training data is created and model performance is evaluated, there is an “Ask the Assistant” button. By clicking this button, users can request specific, targeted advice from the assistant related to those particular areas. (2) When users are not inclined to actively seek advice, the assistant can monitor the entire interaction and periodically provide advice to guide them in the right direction (e.g., Assistant: “You have defined specific category names like ‘Carrot’ and ‘Flower,’ but how about using category names like ‘Edible Plants’ and ‘Non-Edible Plants?’”).

All interactions with the assistant are generated by sending prompts to OpenAI API (gpt-4o-2024-05-13 model)<sup>1</sup> (Fig. 2). These prompts include the entire chat history and the user’s training data. The training data, sent as a single image data file per category, includes category names and associated images. This comprehensive inclusion of information ensures that the assistant always has a complete view of the user’s thoughts and development progress when generating advice. The key aspect of the prompts is that they include instruction like: “Support the user in formulating their tasks by proposing specific examples through dialogue.”

## 3 USE CASE SCENARIO

Suppose a user wants a model to determine whether a plant in an image is edible. Initially, s/he has no idea what categories to define and inputs this requirement in the chat part. The system suggests, “How about defining two categories: ‘Edible’ and ‘Non-Edible?’” The user agrees and follows this suggestion. Next, the user uploads several vegetable images to the ‘Edible’ category and some wild plant images to the ‘Non-Edible’ category. The user is unsure whether the uploaded images are suitable as training data. When s/he clicks the “Ask” button to confirm, the assistant responds positively, and s/he proceeds with training the model.

After the user has completed evaluating the model with images of non-edible flowers, the assistant autonomously provides the following advice: “To further refine and test the accuracy, try introducing unexpected categories such as fruit or an animal.” When the user uploads an animal image, the model produces mismatched results since it only has ‘Edible’ and ‘Non-Edible’ categories. The user asks the assistant: “It seems weird for me with ‘Edible’ and ‘Non-Edible’ for a lion image.” The assistant then suggested defining a new category such as “Not a Plant”. The user follows this advice, uploads animal images for the category, and retrain the model.

This way, through chat with the assistant, the user fulfills their needs and creates training data with appropriately defined categories. This guided interaction is distinctive compared to other MLLM-based applications such as chatbots or copilots. Our system provides users with insights that go beyond mere assistance.

## 4 CONCLUSION

We propose an MLLM-based system that guides users in formulating appropriate tasks and creating training data when developing machine learning models interactively. The MLLM assistant can help users translate their vague needs into creating concrete models.

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<sup>1</sup><https://openai.com/index/hello-gpt-4o/>